

Solution for CS5275 Midterm

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Note: Unlike my solution files for the problem sets, these are more straightforward with primarily only the solutions. For the proof problems, I included a few common mistakes made by students, but for most of the questions, I don't have anything else to say other than the solution.

1 True/False questions

Question 1. Let $G = (V, E)$. If we let H to be a graph over the same vertex set V with conductance $\Theta(1)$ and maximum degree $\Theta(1)$, then $\Phi(G, H)$ is within a constant factor of the edge expansion of G .

Answer: True. First, recall the edge expansion and the generalized conductance:

$$h(G) = \min_{0 < |S| \leq n/2} \frac{|E_G(S, V \setminus S)|}{|S|} \quad \text{and} \quad \Phi(G, H) = \min_{|E_H(S, V \setminus S)| > 0} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

In the definition of $\Phi(G, H)$, we may always make $|S| \leq n/2$ by flipping $S \leftarrow V \setminus S$ otherwise. Also, since $\Phi(H) > 0$, it follows H is connected, so $|E_H(S, V \setminus S)| > 0$ as long as $S \neq \emptyset$. So in fact $\Phi(G, H)$ is defined identically if we take over all S satisfying $0 < |S| \leq n/2$.

Therefore, we will now show that $|S|$ and $|E_H(S, V \setminus S)|$ are within constant factor of each other. This means $h(G)$ and $\Phi(G, H)$ are within constant factor for every particular S , so they remain within constant factor by taking the minimum.

In one direction:

$$\begin{aligned} |S| &\leq \text{vol}_H(S) && (H \text{ is connected}) \\ &\leq \frac{|E_H(S, V \setminus S)|}{\Phi(H)} && (\text{def of conductance}) \\ &\leq \frac{|E_H(S, V \setminus S)|}{\Theta(1)} && (\text{property of } H) \end{aligned}$$

In the other direction:

$$\begin{aligned} |E_H(S, V \setminus S)| &\leq \text{vol}_H(S) && (\text{cut contributes to vol}) \\ &\leq \max \text{ degree} \cdot |S| && (\text{max edges per vertex}) \\ &\leq \Theta(1) \cdot |S| && (\text{property of } H) \end{aligned}$$

This proves that $|S|$ and $|E_H(S, V \setminus S)|$ are within constant factor of each other, so the result follows.

Question 2. For any cycle graph, it is possible to remove at most 0.01 fraction of the edges so that each remaining connected component has conductance $\Omega(1)$.

Answer: True. Remove every 100-th edge. The resulting graph is made of disjoint paths of length ≤ 198 . (Most of them are length exactly 99, but the last piece merges with the first piece.) Since each component is of constant size, the conductance is $\Omega(1)$.

Note that the problem refers only to cycle graphs, not general graphs. I'm not sure about the answer for general graphs – I suspect false – but cycle graphs are much more structured and easier to analyze.

Question 3. For any connected graph G , if the largest eigenvalue of the normalized Laplacian is at least $2(1 - \varepsilon)$, then G is $O(\varepsilon)$ -close to bipartite.

Answer: False. Let $n = 2k$. Consider the graph G defined as follows. Take a k -vertex clique C , and label one of its vertices v_0 . Attach a k -long path to v_0 , and label the vertices $v_1, v_2, \dots, v_k$ from closest to v_0 to farthest.

First, we consider how close it is to being bipartite. It's very far away. We need to turn the clique C bipartite, so the k vertices must be able to be divided into X and Y such that there are no edges inside X and no edges inside Y . This requires removing $\Theta(|X|^2 + |Y|^2)$ edges. Since $|X| + |Y| = k$, by the QM-AM inequality, we need to remove $\Theta(k^2)$ edges.

Now we compute λ_n . Using the hint:

$$\lambda_n = 2 - \min_{\mathbf{x} \neq \mathbf{0}} \frac{\sum_{uv \in E} (x_u + x_v)^2}{\sum_v (\deg(v) \cdot x_v^2)}$$

Now define $\mathbf{x}$ as follows:

$$x_v = \begin{cases} 0 & \text{if } v \text{ is in the clique } C \\ (-1)^k k & \text{if } v = v_k \text{ in the path} \end{cases}$$

Consider λ_n for this $\mathbf{x}$.

For the numerator, every edge in C contributes $(0+0)^2 = 0$. Every edge in the path satisfies $(x_u + x_v)^2 = 1$ by the construction. So the numerator is exactly k .

For the denominator, the clique contributes nothing, but the path contributes

$$\sum_k (\deg(v_k) \cdot x_v^2) \geq \sum_k k^2 = \Theta(k^3).$$

Therefore

$$\lambda_n \geq 2 - \frac{k}{\Theta(k^3)} = 2 - \Theta\left(\frac{1}{k^2}\right)$$

so in this case $\varepsilon = \Theta(1/k^2)$. But we need to remove $\Theta(k^2)$ edges, which is $\Theta(1)$ proportion of all the edges in the graph. This is not $O(\varepsilon)$, so this is a counterexample.

Note: Originally, this problem did not have the hint. But on the slides, you have only learned a formula for λ_n for d -regular graphs, so you would have to find a regular graph counterexample. It exists but it's very ugly.

Question 4. $\Phi(G) \in \Omega(1)$ implies $T_{\text{mix}}(G, 0.001) \in O(1)$.

Answer: False. Choose G that is d -regular for some $d = O(1)$, for example by taking the deterministic construction in lecture 1 slide 35. Consider a distribution supported only on one vertex. Then in $O(1)$ time, the random walk can only reach $O(1)$ many vertices. Thus for the remaining $n - O(1)$ vertices, the probability the random walk is there is 0. But in a regular graph, the stationary probability is $1/n$ for each vertex. Thus the total variation distance is at least

$$(n - O(1)) \cdot \frac{1}{n} = 1 - O\left(\frac{1}{n}\right),$$

which is definitely > 0.001 for large enough n .

Question 5. The transition matrix of the standard (non-lazy) random walk is symmetric even if the graph is not regular.

Answer: False. Let $\mathbf{W}$ be the transition matrix. If u, v are adjacent, then $\mathbf{W}_{uv} = 1/\deg(u)$ but $\mathbf{W}_{vu} = 1/\deg(v)$. So if the graph is not regular, we may have adjacent vertices with different degrees, which makes $\mathbf{W}$ not symmetric.

Question 6. The diameter of any n -vertex Ramanujan graph is $O(\log n)$.

Answer: True. Ramanujan graphs are $\Omega(1)$ -expanders. (Lecture 8 slide 15) Such graphs have $O(\log n)$ diameter. (Lecture 1 slide 24)

Question 7. Let $G = (V, E)$ be any n -vertex graph. Sample k unordered vertex pairs $\{s_1, t_1\}, \{s_2, t_2\}, \dots, \{s_k, t_k\}$, each chosen independently uniformly at random from all possible $\binom{n}{2}$ pairs. There exists some $k = O(n \log n)$ such that, with probability at least $1 - \frac{1}{n}$, there exists an index i such that

$$\kappa_G(s_i, t_i) = \kappa(G).$$

Answer: False. Let G be a clique minus 1 edge uv . Then $\kappa(G) = n - 2$, but this is only achieved by a single pair of vertices (u, v) , so our samples must find this pair. If we only sample $O(n \log n)$ vertices, the probability of missing this pair is

$$\left(1 - \frac{1}{\binom{n}{2}}\right)^{O(n \log n)} \approx \exp\left(-\frac{O(n \log n)}{n}\right) \rightarrow \exp(0) = 1.$$

So we are actually very likely to miss the pair rather than find the pair.

We need somewhere around $\Omega(n^2 \log n)$ samples to have the desired probability of finding the pair.

Question 8. Let $G = (V, E)$ be any graph whose vertex connectivity is $\kappa(G) = k$. Let $S \subseteq V$ be an arbitrary set of k vertices. Construct a graph G' by adding a new vertex r that is adjacent to every vertex in S . Then we must have $\kappa(G') = k$.

Answer: True. First, $\kappa(G') \leq k$. To see this, take any $v \notin S$; since $|S| = \kappa(G) < |V|$, there must exist at least one. Then between r and v in G' , we have the cut $(\{r\}, S, V \setminus S)$ with cut size k .

To show $\kappa(G') \geq k$, suppose otherwise, so we have a cut (L, C, R) in G' with $|C| < k$. Since $|S| = k$, it follows S is not wholly contained in C . So suppose $S \cap L \neq \emptyset$.

If $r \in C$, then we can remove r to obtain a cut $(L, C \setminus \{r\}, R)$ in G . But then $|C \setminus \{r\}| = |C| - 1 < k = \kappa(G)$, contradicting κ .

Otherwise, we cannot have $r \in R$ as then there is an edge between L (some vertex in S) and R (i.e. r). So we have $r \in L$. It also follows $S \cap R = \emptyset$ for a similar reason.

In that case, we can remove r to obtain a cut $(L \setminus \{r\}, C, R)$ in G . Notably, $L \setminus \{r\}$ is not empty, so it is a valid cut. But then $|C| < k = \kappa(G)$, contradicting κ .

So there is no such cut (L, C, R) , so $\kappa(G') \geq k$. This proves $\kappa(G') = k$.

Question 9. The uniform distribution is a stationary distribution for the α -lazy walk.

Answer: True. A stationary distribution satisfies

$$(\mathbf{I} - (1 - \alpha)\mathbf{N})\mathbf{x} = \mathbf{x} \quad \implies \quad (1 - \alpha)\mathbf{N}\mathbf{x} = \mathbf{0}$$

But as long as $\alpha \neq 1$, this equation is the same $\mathbf{N}\mathbf{x} = \mathbf{0}$ no matter what. We know the uniform distribution is a solution for $\mathbf{x}$, so the uniform distribution is a stationary distribution for the α -lazy walk.

Question 10. The α -lazy walk converges to a stationary distribution.

Answer: True. Basically you follow the exact same analysis in lecture 7 slides 13-18, except replacing $\mathbf{I} - \frac{1}{2}\mathbf{N}$ with $\mathbf{I} - (1-\alpha)\mathbf{N}$. Everything else is modified similarly. The summation now uses eigenvalues $1 - (1-\alpha)\lambda_i$. The range of the eigenvalues is now $[1 - 2(1-\alpha), 1]$. All the eigenvalue powers still go to 0 other than the one corresponding to λ_1 , it just may take a longer time.

Note: $\alpha = 0$ corresponds to the standard non-lazy walk. In this case, the range of the eigenvalues is $[-1, 1]$, so an eigenvalue corresponding to -1 will not vanish; indeed, this corresponds to a bipartite component. Meanwhile, $\alpha = 1$ corresponds to a “walk” that never goes anywhere. The range of the eigenvalues is $[1, 1]$, since the transition matrix reduces to just $\mathbf{I}$, and every distribution is a stationary distribution.

Question 11. The sequence of Rayleigh quotients $\{\rho_t\}_{t \geq 0}$ always converges.

Answer: True. We follow the power method in lecture 6. Let $\mathbf{M}$ have eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ with their associated orthonormal eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, and suppose that $|\lambda_1| \leq |\lambda_2| \leq |\lambda_3| \leq \dots \leq |\lambda_n|$. Now write

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n.$$

Some of these c_i 's may be zero, although not all because $\mathbf{x}_0 \neq \mathbf{0}$. Let m be the largest i such that $c_m \neq 0$.

First, note that $\lambda_m \neq 0$. This is because we can write

$$\mathbf{M}\mathbf{x}_0 = \sum_i c_i \lambda_i \mathbf{v}_i$$

All terms $i > m$ are zero because $c_i = 0$. But if $\lambda_m = 0$, then all terms $i \leq m$ are also zero because $|\lambda_i| \leq |\lambda_m| = 0$. This would mean $\mathbf{M}\mathbf{x}_0 = \mathbf{0}$, contradicting the assumption.

Knowing $\lambda_m \neq 0$, the Rayleigh quotient at time t can be written as

$$\rho_t = \frac{\sum_i c_i^2 \lambda_i^{2t+1}}{\sum_i c_i^2 \lambda_i^{2t}} = \lambda_m \cdot \frac{\sum_i c_i^2 (\lambda_i/\lambda_m)^{2t+1}}{\sum_i c_i^2 (\lambda_i/\lambda_m)^{2t}}$$

These sums both go only up to $i = m$, because for all $i > m$, we have $c_i = 0$. Therefore, $|\lambda_i/\lambda_m| \leq 1$ for all i .

If $|\lambda_i/\lambda_m| < 1$ with a strict inequality, then as $t \rightarrow \infty$, we have $(\lambda_i/\lambda_m)^{2t} \rightarrow 0$. Therefore the term vanishes. The only terms left are those where $|\lambda_i| = |\lambda_m|$; this includes $i = m$ itself, and perhaps other eigenvalues at the same magnitude. So the Rayleigh quotient tends to

$$\rho_\infty = \lambda_m \cdot \frac{\sum_i c_i^2 \cdot (\lambda_i/\lambda_m) \cdot 1}{\sum_i c_i^2 \cdot 1}$$

where the summations are over i with $|\lambda_i/\lambda_m| = 1$. This proves convergence.

Question 12. The sequence of normalized vectors $\{\mathbf{y}_t\}_{t \geq 0}$ always converges.

Answer: False. There's a very simple counterexample. Consider

$$\mathbf{M} = [-1] \quad \text{and} \quad \mathbf{x} = (1).$$

Yes, just working in 1 dimension, with a 1×1 matrix. Now

$$\mathbf{M}\mathbf{x} = -\mathbf{x} \quad \implies \quad \mathbf{y}_{t+1} = -\mathbf{y}_t.$$

So the sequence $(\mathbf{y}_t)$ does not converge because it oscillates between two values.

Intuitively, this happens because there's an eigenvalue (here $\lambda_1 = -1$) that is negative and dominates the Rayleigh quotient. So even though we can write $\mathbf{x} = \sum_i c_i \mathbf{v}_i$ and each $\mathbf{v}_i$ is only scaled by λ_i , it might flip direction. The normalized vector $\mathbf{y}$ captures the magnitude but not the direction flip, which is why it does not converge. If the eigenvalues are all nonnegative, it's impossible to flip the direction.

Question 13. If $\Phi(G) \geq \phi$, then G is $O(1/\phi)$ -robust against edge deletion.

Answer: True. Let D be any edge set, and consider the remaining components. For each small component S , we have

$$\text{vol}(S) \leq \frac{1}{\phi} \cdot |E(S, V \setminus S)|$$

by definition of conductance. Therefore, adding up everything,

$$\sum_{\text{small } S} \text{vol}(S) \leq \frac{1}{\phi} \cdot \sum_{\text{small } S} |E(S, V \setminus S)|$$

However, every edge in some $E(S, V \setminus S)$ must belong to D . Furthermore, every edge in D counts for at most two of the $E(S, V \setminus S)$, one for each endpoint. Therefore

$$\sum_{\text{small } S} \text{vol}(S) \leq \frac{1}{\phi} \cdot \sum_{\text{small } S} |E(S, V \setminus S)| \leq \frac{2}{\phi} \cdot |D|$$

This proves the graph is $(2/\phi)$ -robust.

Question 14. If G is C -robust against edge deletion, then $\Phi(G) \in \Omega(1/C)$.

Answer: True. Take any cut $(S, V \setminus S)$ and suppose S is the smaller side, and let D be the set of edges in the cut. By the robustness property, we have

$$\text{vol}(S) \leq C \cdot |E(S, V \setminus S)|$$

Therefore the conductance for each S is $\geq 1/C$, so the conductance overall is $\Omega(1/C)$.

There is one subtle issue: it's possible the cut has $\text{vol}(S) = \text{vol}(V \setminus S) = \text{vol}(V)/2$ exactly. In this case the summation counts both $\text{vol}(S) + \text{vol}(V \setminus S) = \text{vol}(V)$. But in this case, $\text{vol}(S) = \text{vol}(V)/2$, so the conductance is $1/(2C)$, which is still $\Omega(1/C)$.

2 Multiple choice questions

Question 1. What is the conductance of an n -clique?

Answer: C ($\frac{1}{2} + \Theta(n)$). Consider a cut $(S, V \setminus S)$ and suppose S is the smaller side, and let $|S| = k$. Then

$$|E(S, V \setminus S)| = |S| \cdot |V \setminus S| = k(n - k) \quad \text{and} \quad \text{vol}(S) = \sum_{v \in S} \deg(v) = k(n - 1).$$

So the conductance is

$$\Phi(S) = \frac{|E(S, V \setminus S)|}{\text{vol}(S)} = \frac{k(n - k)}{k(n - 1)} = \frac{n - k}{n - 1}$$

The conductance is minimized when k is as large as possible. But because S is the smaller side, we have $k = |S| \leq n/2$. Therefore, depending on the parity of n ,

$$\Phi(G) = \min_{k \leq n/2} \frac{n - k}{n - 1} = \begin{cases} \frac{n/2}{n - 1} = \frac{1}{2} + \frac{1}{2(n - 1)} & \text{if } n \text{ is even} \\ \frac{(n + 1)/2}{n - 1} = \frac{1}{2} + \frac{1}{n - 1} & \text{if } n \text{ is odd} \end{cases}$$

In both cases, $\Phi(G) = 1/2 + \Theta(1/n)$.

Note that, yes, a clique doesn't have "perfect" conductance. (Among connected graphs, only the star graphs have conductance 1.) Also beware calculating $\text{vol}(S)$. Volume is in the entire graph, not just in S , so it's not something like $\binom{k}{2}$.

Question 2. Suppose $G = (V, E)$ is an n -vertex path. Which choice of sampling probability p ensures that for all x, y ,

$$\mathbb{E}[d_A(x, y)] = \Theta(1) \cdot \text{dist}(x, y) ?$$

Answer: D ($1/n$). To prove that the first three options don't work, use Problem set 1 problem 4a. You can show that, for the path,

$$\mathbb{E}[d_A(x, y)] \leq 2 \cdot \mathbb{E}[f(x, A)] = O\left(\frac{1}{p}\right).$$

But if we take x, y to be the ends of the path, we have $\text{dist}(x, y) = \Omega(n)$, so we need $\mathbb{E}[d_A(x, y)] = \Omega(n)$. This can only happen for $p = O(1/n)$, so the first three options don't work.

But how to prove that the last option works? After all, there is a possibility that no p exists, in which case the answer would become "none of the above".

A direct calculation is messy, but we can leverage a trick. With $p = 1/n$, it's likely that A has only one element. And if A only has one element, it's easy to compute $f(x, A)$ and $f(y, A)$, and hence $d_A(x, y)$.

So, we first compute the probability of the event E of $|A| = 1$:

$$\Pr(E) = \binom{n}{1} \cdot \frac{1}{n} \cdot \left(1 - \frac{1}{n}\right)^{n-1} \geq \exp(-1)$$

Now, $d_A(x, y) \geq 0$ for all A , so we split the expectation into whether E happens or not:

$$\begin{aligned} \mathbb{E}[d_A(x, y)] &= \Pr(E) \cdot \mathbb{E}[d_A(x, y)|E] + \Pr(\bar{E}) \cdot \mathbb{E}[d_A(x, y)|\bar{E}] \\ &\geq \exp(-1) \cdot \mathbb{E}[d_A(x, y)|E] + \Pr(\bar{E}) \cdot 0 \\ &= \exp(-1) \cdot \mathbb{E}[d_A(x, y)|E] \end{aligned}$$

Now we are conditioning on E happening, i.e. A contains exactly one vertex v . Divide the path between x, y into three thirds. Observe that, if v does *not* fall into the middle third – it might be in one of the other thirds, or it might be outside the path x, y entirely – then $d_A(x, y) \geq \text{dist}(x, y)/3$. But the probability of v falling into the middle third is $\leq 1/3$. Therefore, if we let the event F to be v falling in the middle third, we get:

$$\begin{aligned} \mathbb{E}[d_A(x, y)|E] &= \Pr(F) \cdot \mathbb{E}[d_A(x, y)|E, F] + \Pr(\bar{F}) \cdot \mathbb{E}[d_A(x, y)|E, \bar{F}] \\ &\geq \Pr(F) \cdot 0 + \frac{2}{3} \cdot \frac{\text{dist}(x, y)}{3} \\ &= \frac{2}{9} \cdot \text{dist}(x, y) \end{aligned}$$

Therefore,

$$\mathbb{E}[d_A(x, y)] \geq \frac{2}{9} \cdot \exp(-1) \cdot \text{dist}(x, y) = \Omega(1) \cdot \text{dist}(x, y).$$

We also have $d_A(x, y) \leq \text{dist}(x, y)$ no matter what, so the embedding has constant distortion. This shows $p = 1/n$ works, and so is the answer.

Question 3. Suppose $G = (V, E)$ is an n -vertex clique. Which choice of sampling probability p ensures that for all x, y ,

$$\mathbb{E}[d_A(x, y)] = \Theta(1) \cdot \text{dist}(x, y) ?$$

Answer: A ($1/2$). The Bourgain embedding for a clique is much easier to compute. We have $d_A(x, y) = 1$ if and only if exactly one of x, y is in A ; it's 0 otherwise. So

$$\mathbb{E}[d_A(x, y)] = \Pr(\text{exactly one of } x, y \text{ is in } A) = 2p(1 - p).$$

Now $\text{dist}(x, y) = 1$, so we need

$$2p(1 - p) = \Theta(1).$$

This only happens if $p, 1 - p$ are both $\Omega(1)$. Among the options, only $p = 1/2$ works. And for $p = 1/2$, we have $\mathbb{E}[d_A(x, y)] = 1/2 = \Theta(1) \cdot \text{dist}(x, y)$ indeed.

Question 4. What is the value of $\sum_{i=1}^n \lambda_i$ on an n -vertex tree?

Answer: D (0). The hint helps a lot, because the trace $\text{tr}(\mathbf{A})$ is a much easier quantity to compute than the eigenvalues. But for a simple graph with no self-loops, the diagonal entries of $\mathbf{A}$ are all zero. So $\text{tr}(\mathbf{A}) = 0$.

Question 5. What is the value of $\sum_{i=1}^n \lambda_i^2$ on an n -vertex tree?

Answer: A ($2(n-1)$). Note that the eigenvalues of $\mathbf{A}^k$ are $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ (e.g. lecture 6 slide 6). So it's enough to compute $\text{tr}(\mathbf{A}^2)$.

What is $\mathbf{A}^2$? We have $\mathbf{A}_{xz}^2 = \sum_y \mathbf{A}_{xy} \mathbf{A}_{yz}$, so this counts the number of walks from x to z of exactly 2 steps. The trace is the sum over the diagonal elements, i.e. $x = z$, so we're counting the number of 2-step walks that end on the same vertex.

Such walks must traverse an edge and then walk back the same edge. Each edge contributes 2 such walks (it can start from either end). On a tree, there are $n - 1$ edges total. So the number of walks is $2(n - 1)$.

Question 6. What is the value of $\sum_{i=1}^n \lambda_i^3$ on an n -vertex tree?

Answer: D (0). Using the same analysis in Question 5, we're looking for $\text{tr}(\mathbf{A}^3)$, and this is the number of 3-step walks that end on the same vertex. In other words, it counts the number of triangles in the graph, multiplied by 6 (because there are 3 ways to choose the starting vertex and 2 ways to choose the direction of the triangle).

But a tree is bipartite, so it has no triangles at all. So there are 0 such walks.

3 Proof problems

3.1 Question 1

Let G be any n -vertex d -regular graph, and let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of its normalized Laplacian $\mathbf{N}$.

- (a) Prove that if G has at least k connected bipartite components, then $\lambda_{n-k+1} = 2$.
- (b) Prove that if $\lambda_{n-k+1} = 2$, then G has at least k connected bipartite components.

Part (a). Continue the reasoning on lecture 4 slide 32. Let C be a bipartite component that has parts A and B . Then $\mathbf{x} = \mathbf{1}_A - \mathbf{1}_B$ is an eigenvector with eigenvalue 2. This is true for each C separately, and all the eigenvectors obtained this way are linearly independent, so they each correspond to their own eigenvalues. So if there are k bipartite components, the eigenvalue 2 has multiplicity k . Since 2 is the largest possible eigenvalue of $\mathbf{N}$, it follows these k instances of eigenvalue 2 must be all the largest, so

$$\lambda_{n-k+1} \text{ (} k\text{-th largest eigenvalue)} = 2.$$

Part (b). Follow the reasoning on lecture 4 slide 33, but we will make it more precise. Consider an eigenvector $\mathbf{x}$ with eigenvalue 2; we will characterize how $\mathbf{x}$ exactly looks. We need $x_u + x_v = 0$ for all $uv \in E$, therefore:

- For a bipartite component C , pick a vertex v_C . Once we determine x_{v_C} , everything else in the component is determined.
- For a non-bipartite component C , we have an odd cycle. Traversing this odd cycle, we get $x_v = 0$ for every vertex in this cycle. And this spreads to the entire component, everything in the component must be 0.

Therefore, after choosing v_C for every bipartite component C , the eigenvectors $\mathbf{x}$ are completely determined by the values of x_{v_C} .

If we have k instances of the eigenvector 2, this means we can find k linearly independent such eigenvectors. Therefore, even when restricting the eigenvectors to only the vertices v_C over all C , we still need them to be k linearly independent vectors. Therefore there must be at least k such v_C 's, so there are at least k bipartite components.

Note: A lot of you got part (b) wrong. If you use the reasoning in lecture 4 slide 33, about taking $A = \{v : x_v > 0\}$ and $B = \{v : x_v < 0\}$, then this does not necessarily give the bipartite components. A and B can be unions of bipartite components. Not just that, different $\mathbf{x}$ that are not in the same span might also give the same A, B .

You need a lot extra care if you want to do this approach. For example, you can choose $\mathbf{x}$ with the least number of nonzero components; this guarantees you're getting a single bipartite component (but you have to prove it). Or you can make sure to choose orthogonal $\mathbf{x}$. In this case, A, B may still be unions of bipartite components, but at least different $\mathbf{x}$ will give different A, B , and you might be able to extract the actual components from there.

But I think it's the easiest to simply try characterizing $\mathbf{x}$ as a whole, like in the solution above.

3.2 Question 2

Prove that, if $n \geq 100 \cdot d$, then any clique in any n -vertex d -regular Ramanujan graph has at most $O(\sqrt{d})$ vertices.

Solution. Use the expander mixing lemma with $S = T =$ the clique. Recalling $\sigma_2 \leq 2\sqrt{d-1}$ for Ramanujan graphs, we have

$$e(S, S) - \frac{d}{n} \cdot |S|^2 \leq (2\sqrt{d-1}) \cdot |S|$$

But also, $e(S, S) = |S|(|S| - 1)$ because S is a clique. Therefore, rearranging,

$$\begin{aligned} |S|^2 - |S| - \frac{d}{n} \cdot |S|^2 &\leq (2\sqrt{d-1}) \cdot |S| \\ \left(1 - \frac{d}{n}\right) \cdot |S|^2 &\leq (1 + 2\sqrt{d-1}) \cdot |S| \\ |S| &\leq \frac{1 + 2\sqrt{d-1}}{1 - d/n} \end{aligned}$$

Since $d/n \leq 0.01$, the denominator is $\Omega(1)$. So $|S| = O(\sqrt{d})$ and we're done.

Note: Somehow a lot of you tried to use the property “if $|S| \cdot |T| \cdot d \geq 4n^2$, then $e(S, T) > 0$ ”.

First, you want to test for a clique, so the entire C is full of edges, not just it's non-empty. What are your S, T ? Why would you want $e(S, T) > 0$?

Some students chose $S =$ the clique C , and $T =$ vertices not adjacent to C , to try to use the contrapositive, knowing that S is full of edges means T is larger than usual. Unfortunately I believe this will not work either; S is just too small that the premise ends up being always false.